

Introduction to Sets and Methods of Describing Sets

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A **set** is a well-defined collection of objects called **elements**.

Elements of a Set

An **element** is an object that belongs to a set.
The symbol $\in$ means “is an element of,” while
 $\notin$ means “is not an element of.”

Roster Form

A set may be written in **roster form**, where all elements are listed inside braces.

$$A = \{2, 4, 6, 8, 10\}$$

Rule Form

A set may also be written in **rule form** (or set-builder form), where a rule describes the elements of the set.

$$A = \{x \mid x \text{ is an even whole number less than } 12\}$$

Empty Set

The **empty set** has no elements and is written as

$\emptyset$

or

$\{\}$

Examples

Example 1

Write the set of even whole numbers less than 12 in roster form and rule form.

Example 2

Let $B = \{m, a, t, h\}$. Determine whether each statement is true or false.

$$a \in B$$

$$x \in B$$

Example 3

Let $D = \{1, 3, 5, 7, 9\}$. Complete each statement using $\in$ or $\notin$.

7 _____ D

4 _____ D

Practice Exercises

Practice 1

Write the set of even whole numbers less than 12 in **roster form** and **rule form**.

Practice 2

Write the set of multiples of 4 from 4 to 24 in **roster** and **rule form**.

Practice 3

Let $B = \{m, a, t, h\}$. Determine whether each statement is true or false.

$$a \in B$$

$$x \in B$$

$$m \notin B$$

$$h \in B$$

Practice 4

Write the set of odd numbers greater than 5 but less than 16 in both roster and rule form.

Practice 5

Write the set of months of the year that begin with the letter “J” in both roster and rule form.

Practice 6

Let $D = \{1, 3, 5, 7, 9\}$. Complete each statement using $\in$ or $\notin$.

$$7 \text{ ______ } D$$

$$4 \text{ ______ } D$$

$$1 \text{ ______ } D$$

$$8 \text{ ______ } D$$

Activities

Activity 1

Write the set of odd whole numbers less than 14 in **roster form** and **rule form**.

Activity 2

Write the set of multiples of 5 from 5 to 30 in **roster** and **rule form**.

Activity 3

Let $F = \{c, l, a, s\}$. Determine whether each statement is true or false.

$$l \in F$$

$$b \in F$$

$$s \notin F$$

$$a \in F$$

Activity 4

Write the set of even numbers greater than 8 but less than 21 in both roster and rule form.

Activity 5

Write the set of days of the week that begin with the letter “T” in both roster and rule form.

Activity 6

Let $H = \{12, 14, 16, 18, 20\}$. Complete each statement using $\in$ or $\notin$.

14 _____ H

17 _____ H

20 _____ H

9 _____ H

Thank you!